

CCNA Exam v1.0 (200-301)

1.0 Network Fundamentals

- 1.1 Explain the role and function of network components
- 1.2 Describe characteristics of network topology architectures
- 1.3 Compare physical interface and cabling types
- 1.4 Identify interface and cable issues (collisions, errors, mismatch duplex, and/or speed)
- 1.5 Compare TCP to UDP
- 1.6 Verify IP parameters for Client OS (Windows, Mac OS, Linux)
- 1.7 Explain virtualization fundamentals (virtual machines)

2.0 Network Access

- 2.1 Describe switching concepts
- 2.2 Configure and verify interswitch connectivity[VLAN]
- 2.3 Configure and verify Layer 2 discovery protocols (Cisco Discovery Protocol and LLDP)
- 2.4 Configure and verify (Layer 2/Layer 3) EtherChannel (LACP)
- 2.5 Describe the need for and basic operations of Rapid PVST+ Spanning Tree Protocol and identify basic operations
- 2.6 Compare Cisco Wireless Architectures and AP modes and Describe physical infrastructure connections of WLAN components (AP, WLC, access)
- 2.7 Configure the components of a wireless LAN access for client connectivity using GUI only such as WLAN creation, security settings, QoS profiles, and advanced WLAN settings

3.0 IP Connectivity

- 3.1 Introduction to Routers
- 3.2 Configure and verify IPv4 addressing and subnetting
- 3.3 Determine how a router makes a forwarding decision by default
- 3.4 Configure and verify IPv4 static routing
- 3.5 Introduction to dynamic routing
- 3.6 Configure and verify RIP
- 3.7 Configure and verify EIGRP
- 3.8 Configure and verify single area OSPFv2
- 3.9 Describe the purpose of first hop redundancy protocol
- 3.10 Compare IPv6 address types

4.0 IP Services

- 4.1 Configure and verify inside source NAT using static and pools
- 4.2 Configure and verify NTP operating in a client and server mode
- 4.3 Explain the function of SNMP in network operations
- 4.4 Describe the use of syslog features including facilities and levels
- 4.5 Configure and verify DHCP client and relay
- 4.6 Explain the forwarding per-hop behavior (PHB) for QoS such as classification, marking, queuing, congestion, policing, shaping
- 4.7 Configure network devices for remote access using SSH
- 4.8 Describe the capabilities and function of TFTP/FTP in the network

5.0 Security Fundamentals

- 5.1 Define key security concepts (threats, vulnerabilities, exploits, and mitigation techniques)
- 5.2 Describe security program elements (user awareness, training, and physical access control)
- 5.3 Describe security password policies elements, such as management, complexity, and
- 5.4 password alternatives (multifactor authentication, certificates, and biometrics)
- 5.5 Describe remote access and site-to-site VPNs
- 5.6 Configure and verify access control lists
- 5.7 Configure Layer 2 security features (DHCP snooping, dynamic ARP inspection, and port security)
- 5.8 Differentiate authentication, authorization, and accounting concepts

6.0 Automation and Programmability

- 6.1 Explain how automation impacts network management
- 6.2 Compare traditional networks with controller-based networking
- 6.3 Describe controller-based and software defined architectures (overlay, underlay, and fabric)
- 6.4 Compare traditional campus device management with Cisco DNA Center enabled device management
- 6.5 Recognize the capabilities of configuration management mechanisms Puppet, Chef, and Ansible
- 6.6 Interpret JSON encoded data